

SEA-AD is a multimodal cellular atlas and resource for Alzheimer's disease

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The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) is a multifaceted open-data resource that is designed to identify cellular and molecular pathologies that underlie Alzheimer's disease. Integrating neuropathology, single-cell and spatial genomics, and longitudinal clinical metadata, SEA-AD is a unique resource for studying the pathogenesis of Alzheimer's disease and related dementias.

Alzheimer's disease (AD) is one of the most important biomedical challenges of our time and affects one in nine Americans over 65; the prevalence is likely to double or triple by 2060 as the population ages¹. Neuropathologically, AD is a complex etiology that is characterized by the temporally and anatomically dyssynchronous but overlapping progressive deposition of hallmark pathological peptides β -amyloid and hyperphosphorylated tau (pTau), with subsequent neurodegeneration^{2,3}. Notwithstanding decades of efforts, treatment strategies that aim to reduce the burden of these pathological peptides have resulted in – at best – modest effects on pathology and small clinical benefits⁴. One impediment has been our lack of understanding of the normative complexity of the brain's cell types and how they are disrupted in disease. There is a tremendous need to identify a new generation of diagnostic markers and therapeutic targets, which necessitates a markedly improved understanding of the underlying biology of AD initiation and progression.

The importance of brain-wide cellular and molecular atlases in AD

A promising path toward gaining such understanding has been paved by the National Institutes of Health (NIH) BRAIN Initiative Cell Census Network (BICCN) and Cell Atlas Network (BICAN). These efforts have focused on establishing powerful technologies for cellular-level analyses that can be applied in brain tissues of human and animal model organisms. Most notably, single-cell transcriptomics and epigenomics can now be used to comprehensively define and characterize the cellular composition of complex nervous system tissues, and spatial transcriptomics can define their tissue organization and functional local architecture. Applied to specific brain regions, these and associated technologies are creating foundational reference cell classifications at unprecedented resolution (in many ways akin to the Human Genome

Project for genomics), as well as an increasingly deep knowledge of the properties of these cells, their tissue organization and contribution to functional neural circuitry⁵. BICCN has produced brain-wide cell atlases in mouse⁶, and an initial version of a human brain cell atlas⁷ that is now being expanded in the BICAN. Modern cellular-resolution atlases provide new standard references for comparison in AD and other brain diseases at the level of specific cell types, molecular programs and gene regulatory networks.

Standards and protocols developed through BICCN for single-cell data generation, anatomic mapping, and data analysis are producing references that can now be applied in translational research. The SEA-AD project builds on these advanced methods to systematically produce a high-resolution, multimodal, multi-brain-region cell atlas of AD as a catalytic resource for the biomedical research community. This consortium project is based at the Allen Institute for Brain Science, with collaborators at the University of Washington and associated AD Research Center (ADRC) and Kaiser Permanente Washington Health Research Institute including the Adult Changes in Thought (ACT) study. Together this represents a US \$40.7 million investment by the NIH National Institute on Aging running through 2025. There are plans for a second phase to extend through 2030 to include broad anatomical coverage of affected brain regions and joint community analysis, with a large highly diverse cohort. In contrast to large-scale imaging-based atlases in AD⁸, modern molecular methods produce extraordinarily complex data that can describe what is affected with AD progression, where it is happening, and when it occurs as a function of modeled disease severity. Taken together, these data demonstrate that there are robust and consistent disease phenotypes but contradict a simple description of disease mechanism. The goal of SEA-AD is to decipher this complexity to understand the underlying biology of AD progression and present these data effectively to the research community. This approach is essential to facilitating the identification of new cellular and molecular targets for mechanistic studies and clinical interventions.

SEA-AD cohort and study design

Three key principles underlie SEA-AD study design. First, cellular and molecular hallmarks of AD will correlate with the aggregate local burden of neuropathology, which should be measured quantitatively in the specific brain regions analyzed and used to model disease severity. Second, a high degree of cellular granularity is essential, and achieved by sampling enough cells to capture each cell type in every donor. Finally, it is extremely important to understand disease progression across the brain and to model the earliest events when intervention may still be possible. This study design enables a critical examination of cell-type pathology and vulnerabilities in AD and the ability to track these changes over disease course.

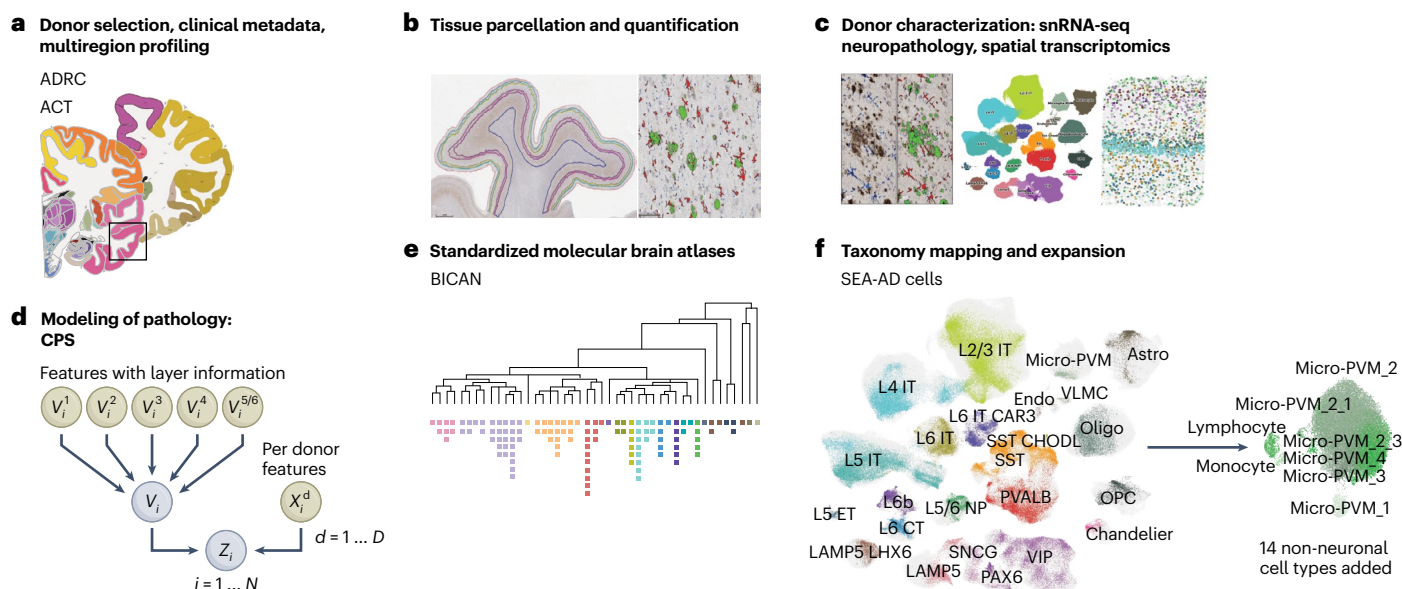


Fig. 1 | SEA-AD project study design. SEA-AD workflow integrates molecular standards, image-based neuropathology, clinical metadata and modeling of disease trajectory to produce a new atlas of AD. **a**, Donor selection through ADRC and ACT enables a multiregion profile across a diverse cohort. **b**, Machine-learning automated tissue parcellation and quantification. **c**, Neuropathology and molecular characterization using single-cell methods. **d**, Statistical modeling of continuous pseudo-time disease severity and progression. **e**, Brain-wide single-cell molecular atlases developed

through BICAN provide a framework for comparison and analysis in AD. **f**, BICAN taxonomy is expanded to include disease-modified cell types for further study. Astro, astrocyte; CPS, continuous pseudo-progression score; CT, corticothalamic; endo, endothelial; ET, extratentorial; IT, intratentorial; micro, microglia; oligo, oligodendrocyte; NP, near-projecting; OPC, oligodendrocyte precursor cell; PVM, perivascular macrophage; snRNA-seq, single-nucleus RNA sequencing; VLMC, vascular leptomeningeal cell.

A well-characterized donor cohort that spans the full spectrum of AD pathology is essential to address the SEA-AD design principles. The initial cohort is composed of 84 individuals who span the full spectrum of AD pathology, including donors with no pathology and those with both AD pathology and common comorbidities (age at death ranges from 65 to 102 years (mean = 88); female/male individuals = 51/33, consistent with the known prevalence of AD in female individuals; and dementia/no dementia = 42/42). Diversity of the participants in age, sex and disease status using BICCN-inspired optimized preparation methods provided valuable high-quality human brain tissues. Extensive donor metadata through the ACT study included cognitive status, phenotypic biomarkers and longitudinal medical history.

SEA-AD first profiled the middle temporal gyrus (MTG), an area that is involved in language and semantic memory processing⁹ and higher-order visual processing¹⁰ (Fig. 1a). From the seminal histopathology of Braak and Braak¹¹ to modern longitudinal characterization of tau by positron emission tomography (PET) imaging¹², numerous studies suggest that the MTG is a transition zone between aging or preclinical AD-related medial temporal lobe or limbic pTau and more advanced stages of AD in which neocortical pTau extends across the brain and is correlated with dementia. The MTG is also the human cortical region with the best-annotated BICCN cell classification, including cellular phenotype data (morphological and physiological properties), because this region is frequently resected for treatment of intractable epilepsy and is used to study cellular phenotypes in living human tissues. Data are also available for the dorsolateral prefrontal cortex (DLPFC) and present ongoing efforts are focusing on other cortical and subcortical regions affected over AD progression.

To analyze cellular pathology in tissue, automated image analysis and deep-learning-based cell segmentation provide measures of quantitative neuropathology burden (Fig. 1b) that enable visual and quantitative analysis of key protein markers such as β -amyloid, pTau and α -synuclein. Multimodal molecular characterization includes quantitative image-based immunohistochemistry, single-nucleus transcriptomics and epigenomics, and targeted spatial transcriptomics (Fig. 1c), which provides a comprehensive view of cell type. A continuous pseudo-progression score derived from the quantitative neuropathology measurements was constructed to model disease severity as a function of neuropathological burden, which ordered donors along a neuropathological continuum and increased discovery power to identify molecular and cellular differences (Fig. 1d). This model connects the quantitative neuropathology of Fig. 1b with donor metadata and disease severity and burden. Single-nucleus RNA-sequencing data are then mapped to the BICCN neurotypical reference cell classification and used to expand that classification to include disease cell states (Fig. 1f). Combining these approaches results in an integrative framework for studying molecular and cell type processes in AD and their connection with the unaffected or normative brain.

SEA-AD open community resource

The SEA-AD online portal is an open-access community resource that is designed to make these data publicly available with quantitative analysis and visualization tools for AD research¹³ (Fig. 2, SEA-AD.org). The portal hosts a suite of tools, based on the techniques described above and ranging from highly interactive to direct data download. Our objective is to make data useful to diverse groups of users, including

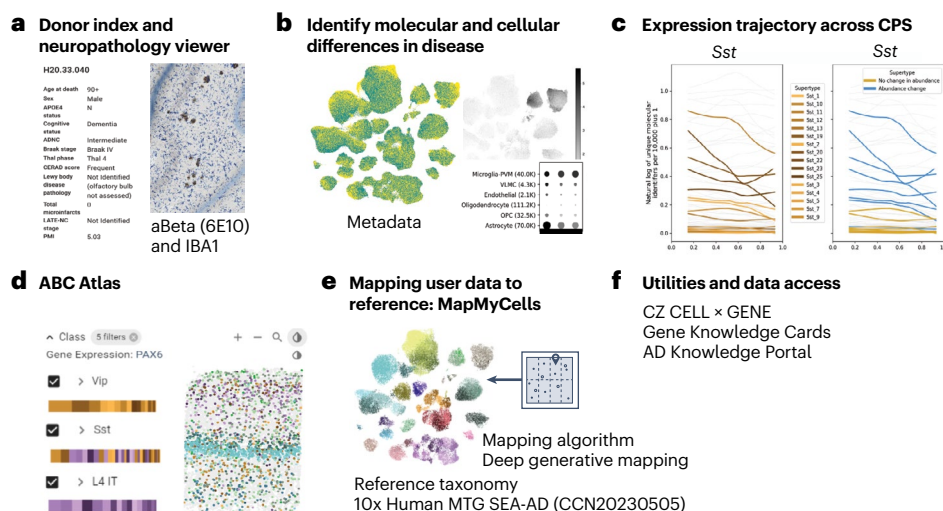


Fig. 2 | SEA-AD data portal and resources. **a–f**, Resources and tools available through the SEA-AD portal enable users to: view donor metadata and neuropathology images (**a**); perform comparative analysis of molecular differences in disease state (**b**); investigate molecular changes within cell types with respect to neuropathology and pseudotime progression (**c**); integrate

with the Allen Brain Cell Atlas for whole-brain single-cell expression and spatial transcriptomics (**d**); use the MapMyCells tool for mapping user data to the BICCN standard atlas in the MTG (**e**); have access to data, summaries and documentation with links to external resources through [Sage Bionetworks AD Knowledge Portal](#), [Open Data on AWS](#) and [SEA-AD.org](#) (**f**).

researchers, clinicians, students and others with less specialized backgrounds.

Core resource features include a [donor index](#) that presents demographic, clinical, cognitive and neuropathological information, and a [neuropathology image viewer](#) (Fig. 2a) that presents zoomable and pannable images of donor brain tissue sections from multiple brain regions stained for key pathological proteins and cell types of interest to AD, including image analysis masks for signal detection. A [transcriptomics comparative viewer](#) enables side-by-side comparison of gene expression for any gene in the neurotypical reference and SEA-AD donors, comparison with essential donor metadata, and quantification of expression differences in AD data in MTG (Fig. 2b). The viewer allows users to change hierarchical levels in the taxonomy for increasing cell-type resolution. To connect neuropathology with the transcriptomic data, the trajectory of differences in gene expression across continuous pseudo-progression scores can be viewed using the [Gene Expression Trajectory Viewer](#) (Fig. 2c). Close integration with the [Allen Brain Cell \(ABC\) Atlas](#), a primary online resource for single-cell molecular data, enables users to study spatial transcriptomics (multiplexed error-robust fluorescence in situ hybridization (MERFISH)) data from SEA-AD donors who span the spectrum of AD pathology in relation to single-nucleus data from the same donors (Fig. 2d).

The SEA-AD portal also provides a tool, [MapMyCells](#), that enables researchers to map their single-cell or single-nucleus transcriptomics datasets to the highly annotated reference taxonomies developed through BICCN, BICAN and the SEA-AD consortia, and automatically assign cell identities using state-of-the-art machine-learning mapping algorithms (Fig. 2e). Best-in-class third-party tools add key additional functionality for visualizing and annotating SEA-AD data (Fig. 2f), such as linkage to the [UCSC genome browser](#) for single-nucleus assay for transposase-accessible chromatin using sequencing (ATAC-seq) epigenetic data, and visualization and annotation through the Chan Zuckerberg Initiative [CELLxGENE tool](#). SEA-AD data and documentation are

made publicly available through the [SEA-AD portal](#), [Sage Bionetworks AD Knowledge Portal](#)¹⁴ and [Open Data on AWS](#) (Amazon Web Services).

Initial findings from SEA-AD

The first major analysis of the SEA-AD datasets illustrates the value of this integrated approach and the power of the SEA-AD resources¹⁵. Mapping SEA-AD data to the BICCN reference, which can be done using MapMyCells (Fig. 2e), resulted in a highly granular AD brain cell-type taxonomy, expanded from the neurotypical taxonomy to include novel disease-associated cell types (for example, microglial and astrocytic cell types). Further molecular analysis, as can be explored using the expression trajectory viewer (Fig. 2c), identified highly specific neuronal loss representing selective vulnerability, with different vulnerable cell-type populations affected at different stages of disease progression and severity. Transcriptomic analysis found a subset of somatostatin-positive inhibitory interneurons whose vulnerability was associated with the lowest (earliest) levels of pathology or disease severity, and before the loss of specific excitatory projection neurons. Spatial transcriptomic analysis demonstrated that most of the vulnerable neuron populations are selectively located in upper cortical layers and, by linking to the BICCN reference, the extensive properties of these cells are known. These findings illustrate the power of the SEA-AD approach to map complex molecular and cellular disease-associated changes to highly resolved cell types across the brain. These data and initial results can be explored and built upon by the AD research community using the extensive suite of tools available at [SEA-AD.org](#).

Summary

SEA-AD represents the fruitful application of the BRAIN Initiative's foundational efforts to understand human brain structure and function to gain a deeper understanding of AD biology, and we invite the community to explore this resource. This approach has great promise for the future of the field, as the approach of data integration to the

BICCN reference is highly extensible, and mapping data to a common reference provides a unifying framework for the AD research community that enables direct comparison and validation across studies. Multiple published datasets mapped to the SEA-AD reference have cross-validated selective cell vulnerabilities across datasets¹⁵, and the use of a common language to describe phenotypes of disease across studies should accelerate progress and reproducibility across the field. Through enlarging and diversifying the donor cohort, an expansion to additional early-affected brain regions, integrating existing results from the field, and developing powerful tools for data exploration, SEA-AD aspires to help the community to provide a detailed foundational understanding of AD biology that will catalyze new discoveries and paths to therapeutic intervention.

Data availability

Access to all data and tools are available via the SEA-AD consortium's web portal at [SEA-AD.org](https://sea-ad.org). Sequencing data are available through controlled access at Sage Bionetworks (accession [syn26223298](https://synapse.org/syn26223298)). [Instructions for access to data on the AD Knowledge Portal](#) is provided by Sage Bionetworks.

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Competing interests

The authors declare no competing interests.